

Characterizing Visibility Graphs of Polygons

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Abstract

Visibility graphs map out direct line-of-sight connections between the vertices of a polygon, which is useful in areas such as robotics, computer vision, and graph theory. However, characterizing a given graph as a visibility graph is an open problem in computational geometry. Using geometric region approximation methods, we have developed an interactive web application to compute the visibility graph of small polygons.

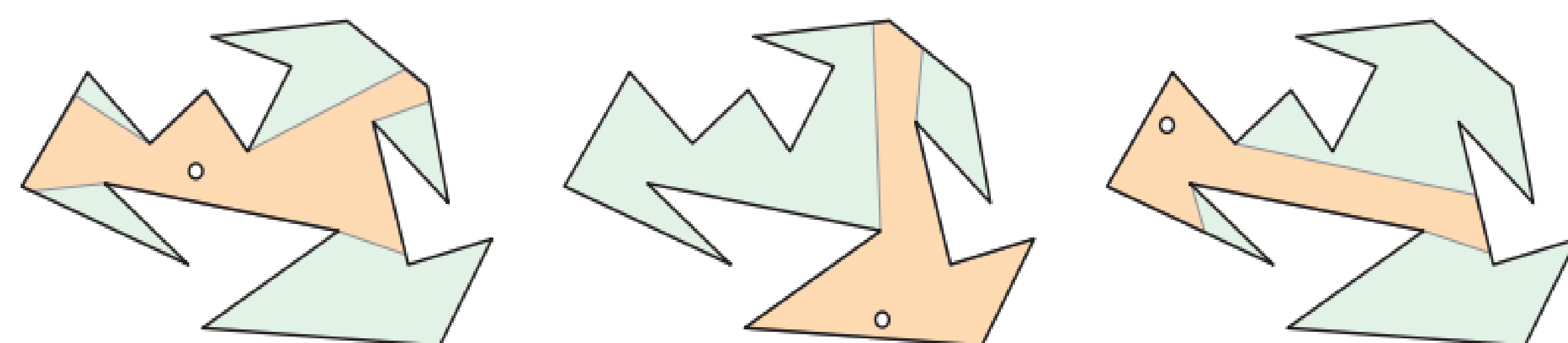
Introduction

Computational geometry is a field at the intersection of mathematics and computer science, which studies algorithms that can be used to solve problems in discrete geometry, especially in low dimensions. The field has applications in robotics, computer graphics, and geographic information systems (GIS). A central theme in this field is to understand spatial relationships such as proximity, intersection, and visibility of geometric objectives relative to each other

The Art Gallery Problem

One of the fundamental problems in discrete geometry is the art gallery problem. We begin by modeling an art gallery as a polygonal region. The key question is: **How many guards are needed to ensure that every point inside the art gallery is visible to at least one guard?**

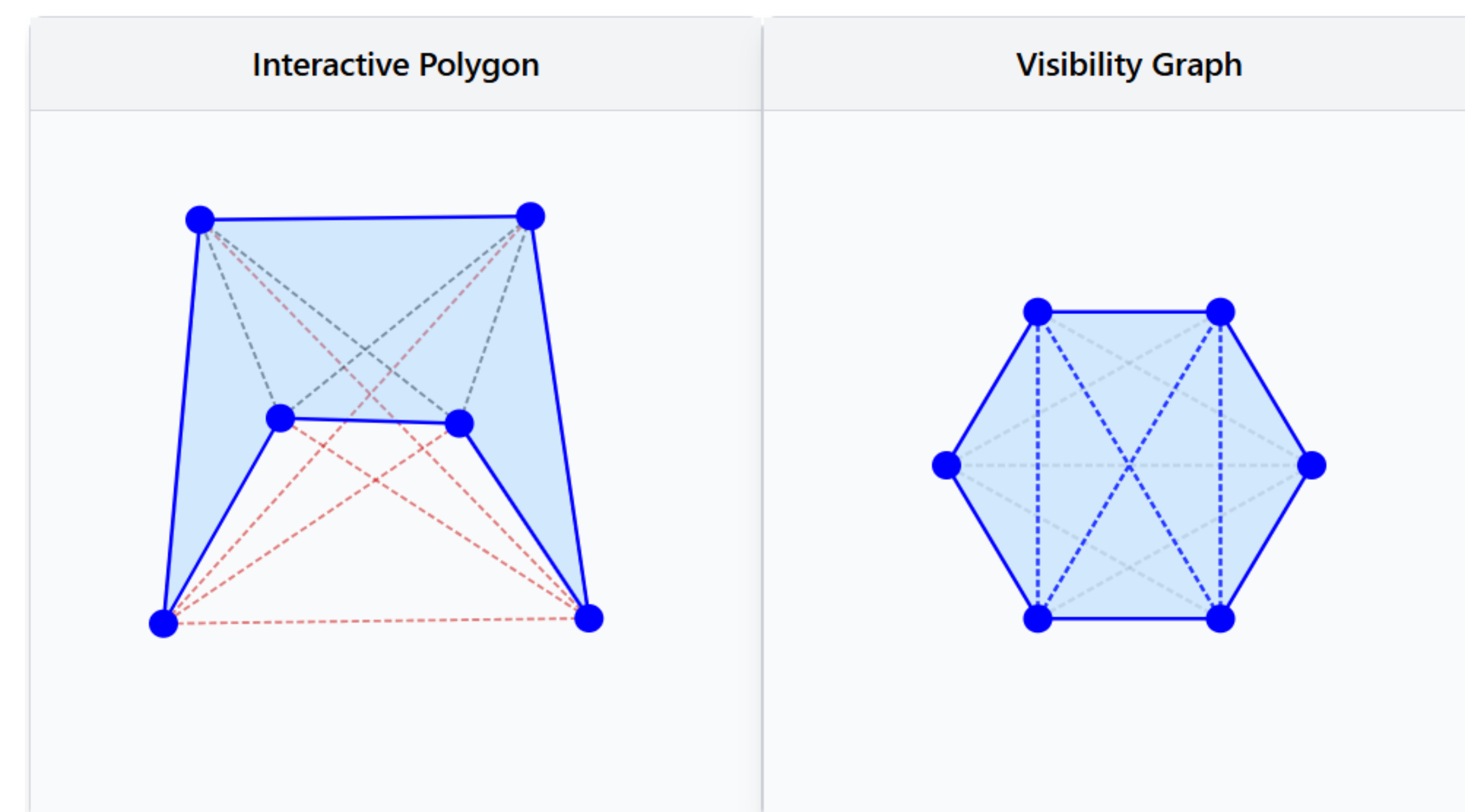
- Basic art gallery theorem: $\lfloor n/3 \rfloor$ guards suffice for a simple polygon with n vertices. (1)
- Many variations of the problem exist: what if the walls were transparent; what if the walls were mirror; and many more.
- Below is an art gallery with 3 distinct guard placements. (1)



The Visibility Graph Problem

The visibility graph of a polygon is a graph representation of which vertices can "see" each other.

- Bijection between graph nodes and polygon vertices. There exists an edge between two nodes if their corresponding vertices can see each other.
- Unsolved problem: **Deciding whether a given graph is the visibility graph of some polygon, and furthermore which sort of polygon it is.**
- At least three are fully known: funnels, spirals, and staircases.
- We made some progress on the characterization of rectangular pipes.



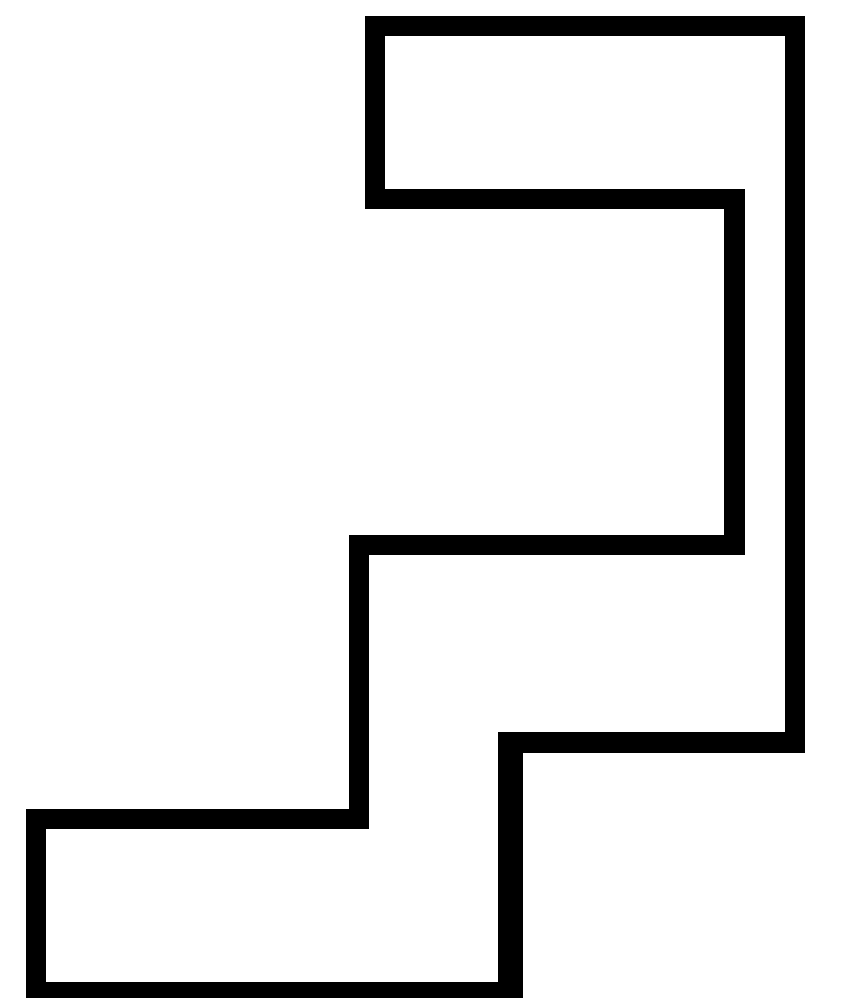
Visibility Graph Visualization Tool

During our research to classify visibility graphs, we identified a gap: there was no existing tool to manipulate a polygon and immediately visualize its visibility graph. To address this, we developed a tool that allows users to manipulate the vertices of a polygon and automatically see a generated visibility graph of that polygon. Our implementation uses geometric region approximation methods to ensure accurate and efficient computation of visibility relations. This tool will be key in future conjecture development for visibility graphs.

Rectangular Pipes

We found some weak results for the visibility graphs of rectangular pipes, based on the results of the Abello et al paper on visibility graphs of staircase polygons (3).

We also proved that there is an injection from the set of visibility graphs of polygons to the set of visibility graphs of polygons with integer coordinate vertices (though it is possible that this theorem was already known), so we can conclude that this technique fails to simplify the problem.



Future Research Directions

Further characterization of polygons: Our tool can be used to extend analysis into new types of polygons (e.g. star-shaped and non-simple) to identify patterns and necessary conditions for a graph to be a valid visibility graph of such a polygon

Another direction is to study problems that maximize or minimize total visibility within a polygonal region. By treating visibility as a measurable quantity, we can explore how changes such as vertex repositioning or adding barriers affect overall visibility. Visibility graphs provide a natural foundation for these studies, capturing pairwise visibility relationships that can be directly analyzed and optimized.



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References

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